

Topic 1

The Generative AI Landscape in Education: Mapping the Terrain of Opportunities, Challenges, and Student Perception

Base Paper Details

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Abstract

Generative Artificial Intelligence (GenAI) has emerged as one of the most influential technologies transforming modern education. Advanced models such as ChatGPT, Gemini, Claude, Microsoft Copilot and image generation systems are enabling intelligent tutoring, adaptive learning, automated content generation, instant feedback, assessment support and personalized educational experiences. This seminar explores the evolution of Generative AI, its underlying concepts, and its growing adoption in schools and higher education. The discussion highlights opportunities such as improved accessibility, personalized learning paths, teacher assistance, multilingual education and enhanced student engagement. It also examines critical challenges including hallucinations, plagiarism, academic integrity, copyright, bias, privacy, security and ethical concerns. The seminar further reviews student perception studies, responsible AI practices and future research directions including AI tutors, Retrieval-Augmented Generation (RAG), multimodal AI and human-AI collaboration. Based on a recent IEEE Access review paper, this seminar provides both technical depth and practical relevance for understanding the future of AI-enabled education.

Topic 2

A Comprehensive Review on Generative AI for Education

Base Paper Details

Authors: Uday Mittal, Siva Sai, Vinay Chamola, Devika

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Abstract

This seminar presents a comprehensive review of Generative Artificial Intelligence and its transformative role in smart education. It introduces the core technologies behind modern GenAI, including Large Language Models (LLMs), Generative Adversarial Networks (GANs), diffusion models and transformer architectures. The seminar explains how these technologies are applied to intelligent tutoring systems, adaptive learning platforms, automated assessment, content generation, virtual laboratories and immersive learning through the metaverse. It also discusses the benefits of AI-assisted teaching, personalized learning, multilingual education and improved accessibility for learners with diverse needs. In addition, the seminar critically analyzes challenges related to privacy, fairness, explainability, accountability, misinformation and ethical AI deployment. Finally, it explores future research opportunities emphasizing responsible AI, human-AI collaboration and next-generation intelligent educational systems capable of delivering engaging, efficient and personalized learning experiences.

Topic 3

A Comprehensive Study on Personalized Learning Recommendation in e-Learning System

Base Paper Details

Authors: Qiu Bin, Megat F. Zuhairi, Jacques Morcos

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Journal: IEEE Access

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Abstract

Personalized learning recommendation systems have become a cornerstone of intelligent e-learning platforms by recommending educational resources according to learner behaviour, interests, learning style and academic progress. This seminar examines the evolution of recommendation systems used in education, covering collaborative filtering, content-based filtering, hybrid recommendation models, machine learning, deep learning, reinforcement learning, graph neural networks and knowledge graph-based approaches. It also explains learner modelling, MOOCs, Learning Management Systems (LMS), adaptive learning environments and educational data mining techniques. Furthermore, the seminar discusses major research challenges including cold-start problems, scalability, explainability, learner privacy and data quality. Future trends such as AI-powered educational assistants, real-time personalization, predictive analytics and intelligent adaptive learning ecosystems are also explored, making this topic highly relevant for the future of digital education.